

User Guide

Step · Hop · Scan

Overview

StationFinder provides three complementary ways to navigate a frequency list loaded from a database file (EiBi, Aoki, HFCC, ILG, or similar). Each method is designed for a different use case and works in conjunction with SDRconnect via WebSocket, other SDR programs via CAT (Kenwood) or with Icom radios over a CI-V serial connection.

Feature	What it does
Step	Manually step through each frequency in the table one at a time, forward or backward.
Hop	Automatically cycle through all frequencies in the table with a configurable dwell time between each.
Scan	Sweep the SDR across a frequency range and find active stations using a fast detection signal analysis technique.

Prerequisites

All three features require the following before they become active:

- An active connection to your SDR receiver (WebSocket to SDRConnect, CAT serial, or CI-V serial).
- A frequency database file successfully loaded (use the Load File button).
- Search mode set to Range (e.g. 9000-10000 kHz). Step, Hop and Scan are only available in Range mode.

| Tip: If any button appears dimmed, check that all three prerequisites above are satisfied.

Step

What Step Does

Step tunes your SDR to each frequency in the results table one at a time. It is a manual, on-demand action — nothing happens automatically. Use Step when you want full control over which frequency to listen to next. The current frequency will be highlighted in green in the table.

How to Use Step

- Left-click the Step button to advance forward to the next frequency in the table.
- Right-click the Step button to step backward to the previous frequency.
- On the first press, Step anchors to the frequency in the table closest to where your SDR is currently tuned, then immediately advances one step.

- Subsequent presses continue forward (or backward) from the current position.

| *Step sends the frequency only — it does not change the demodulation mode.*

Step in Halted Scan State

After a scan completes and is in the Halted state, Step changes behaviour: it only steps through frequencies where the scan detected an active signal. This lets you quickly review only the stations that were found.

When Step is Disabled

- Scan is currently running.
- Search mode is not Range.
- No connection is active.
- The results table is empty.

Hop

What Hop Does

Hop automatically cycles through all frequencies in the results table, tuning your SDR to each one and pausing for a set dwell time before moving to the next. It continues looping until you stop it. Use Hop when you want to monitor a list of frequencies hands-free.

How to Use Hop

- Set the dwell time using the Dwell dropdown (seconds to pause on each frequency).
- Click Hop to start. The button label changes to Stop Hop and turns amber while active.
- The current frequency will be highlighted in green in the table.
- Click Stop Hop to pause. The SDR stays on the current frequency.
- Click Hop again to resume from where it stopped.

| *Hop sends frequency only — it does not change the demodulation mode.*

Hop in Running or Halted Scan State

When a scan is running or has been halted, Hop only cycles through frequencies where the scan detected an active signal. This focuses your monitoring on stations the scan confirmed as active.

In idle mode (no scan), Hop cycles through all visible rows in the table.

Dwell Time

The Dwell dropdown sets how many seconds Hop pauses on each frequency before moving on. Choose a longer dwell if you need time to identify a station; a shorter dwell for rapid monitoring.

When Hop is Disabled

- No connection is active.
- Idle mode with no rows in the table or not in Range mode.

Scan

What Scan Does

Scan uses a FFT (Fast Fourier Transform) technique using IQ data from SDRConnect to detect active signals across a frequency range. The frequency range is determined by the sample rate and the center frequency. This FFT technique is very fast (under 2 seconds for initial detection) and marks frequencies in the results table where a signal was detected above the SNR threshold. SNR averaging is used to maintain the table highlighting during signal fading.

Scan is only available over a WebSocket connection to SDRConnect. CAT and CI-V connections do not support Scan.

Demodulation Mode Requirement

Scan requires SDRConnect to be in AM or SAM demodulation mode. These modes permit carrier detection and SNR analysis that are required for signal detection. USB, LSB, CW, and WFM modes will produce unreliable results and will not work with Scan mode in the current version of StationFinder.

If SDRConnect is not in AM or SAM when you click Scan, the Pre-Scan Review modal will warn you and the Start Scan button will be disabled until you select a valid mode.

Pre-Scan Review Modal

Clicking the Scan button opens the Pre-Scan Review modal before starting. Here you can:

- Review the spectrum display showing the current SDR frequency coverage.
- Adjust the search range bar — drag the centre to shift the range, drag the edges to widen or narrow it.
- Review the Sample Rate, Centre Frequency, SNR Threshold, and database entry count for the selected range. Left-click Sample Rate, Centre Frequency, or SNR to increase the value. Right-click to decrease.
- Select AM or SAM mode using the buttons at the bottom if SDRConnect is not already in a valid mode.
- Click Start Scan to begin (only active when mode is AM or SAM and the range is valid)..

During a Scan

- The Scan button label changes to Stop Scan and turns red while scanning is active.
- The FFT detection algorithm steps across the defined range automatically.
- Rows in the results table are highlighted when a signal is detected at that frequency.
- Changing the SDR centre frequency or sample rate manually in SDRconnect will stop the scan.

Scan States

State	Description
Running	Scan is actively stepping through the range and detecting signals. Hop will operate in this state. Detected (active) stations will be highlighted in green in the table.
Halted	Scan has completed or been stopped. Results are retained and highlighted in the table. Step and Hop will operate in this state
Idle	No scan in progress. Table shows all matching database entries normally.

Clearing Scan Results

While in the Halted state, click the Clear button to dismiss the scan results and return to Idle. This restores the full results table and re-enables normal Step and Hop behaviour.

SNR Threshold

The SNR (Signal-to-Noise Ratio) threshold determines how strong a signal must be relative to the noise floor to be counted as a detection. A higher threshold reduces false positives but may miss weaker stations. Adjust using left-click (increase) and right-click (decrease) on the SNR tile in the Pre-Scan modal. You can also select a range of SNR like 10 to 20 dB.

When Scan is Disabled

- No WebSocket connection is active (CAT/CI-V connections do not support Scan).
 - Search mode is not Range.
 - No database file is loaded.
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Tips & Workflow Suggestions

Recommended Scan Workflow

- Load a database file and enter a frequency range (e.g. 9000-10000 for the 31m band). The easiest way is to use the Band button and select the desired band like 31M. This will issue SDRconnect with an appropriate sample rate, center frequency and AM mode.
- Connect to SDRConnect via WebSocket and ensure the mode is AM or SAM.
- Click Scan, review the Pre-Scan modal, then click Start Scan.
- While Scan is running use Hop to automatically cycle through detected stations. You can halt the Scan and use Hop or Step to change stations..
- Click Clear when finished to return to normal browsing.

Combining Step and Hop

Step and Hop work independently — you can use Step to move to a specific frequency, then start Hop to begin automatic cycling from that position onward.

Filtering Before Hop

Apply language, source, or time filters before starting Hop to narrow the frequency list. Hop will update its list dynamically as filters change, even while running.